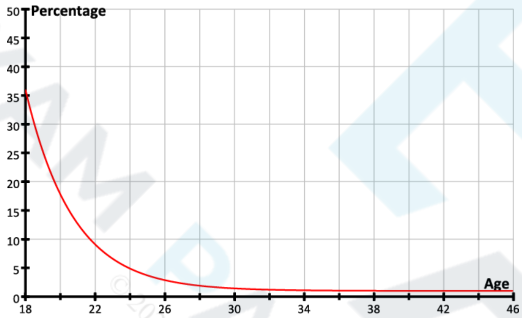
Question			Answer/Indicative content	Marks	Guidance
1			With no more than 3 points on his licence the driver's premium is not altered: £520 The driver now has 3 + 6 = 9 points. The new premium is $£520 \times 2^{\frac{9}{6}} = £1470.78$	B1 B1	£520 £1470.78 or £1471 [but not £1470.80] **Examiner's Comments** Most candidates were successful here. A few did not round a money answer to an appropriate accuracy.
			Total	2	
2		i		B1	A sketch. Must be at least for values of *x* between 18 and 45. Must be correct shape ie descending curve, does not curve up at end, does not cross the horizontal axis (even if extended) ie must → 1. Does not need to go through points exactly as a sketch. **Examiner's Comments** The graph was generally good. The commonest error was that it often stopped at x=44 instead of continuing to at least x=45. In other cases the graph looked as though it would cut through the x axis when produced.

Question			Answer/Indicative content	Marks	Guidance
		ii	For large values of x, $be^{-k(x-17)} \to 0$, and so $y \to a$ which is 1 in this case.	B1	Need x becoming large or $\to \infty$, exponential term $\to$ oe and $y \to a$, $a = 1$ or $y = 1$, $a = 1$ oe NOT just at $x = 45$, $y = 1$ **Examiner's Comments** Few candidates were able to give a complete solution that as $x \to \infty$, $be^{-k(x-17)} \to 0$ and so as $y \to a$, $a = 1$ or equivalent. Many referred to the general shape of the curve or asymptotes or said that at x=45, y=1.Generally the responses were unclear or incomplete.
		iii	Substituting $x = 23 \Rightarrow y = 6.6$	M1	subst $x = 23$ and finding $y = 6.6$ (6.60115…) (or subst $y = 6$ and finding $x = 23.3$ (23.3097…) Only accept comparing x or y, not coefficients
		iii	This is quite close to the observed value of $y = 6$.	A1	(or close to $x = 23$) Accept say, $y = 6.6 \approx 6$. But not say at x = 23, y $\approx$ 6 with no evaluation seen Accept if states, say, $y = 6.6$ which is **not** consistent with $y = 6$ oe **Examiner's Comments** This was well answered although some candidates forgot to compare the result of their calculation with the given point. Most compared 6 with 6.60… but some used 23. A few tried to establish the given constants b and k.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
3			<table><tr><th>Year</th><th>% discount</th></tr><tr><td>2007</td><td>0</td></tr><tr><td>2008</td><td>30</td></tr><tr><td>2009</td><td>40</td></tr><tr><td>2010</td><td>50</td></tr><tr><td>2011</td><td>60</td></tr><tr><td>2012</td><td>65</td></tr><tr><td>2013</td><td>65</td></tr><tr><td>**Total**</td><td>**310**</td></tr></table>	B1	obtaining 310 or 3.1 oe [from adding all relevant terms ie 0 + 30 + 40 + 50 + 60 + 65 + 65 = 310 or 0 + 0.3 + 0.4 + 0.5 + 0.6 + 0.65 + 0.65 = 3.1 soi (with or without first zero term) or from 700 - 100 - 70 - 60 - 50 - 40 - 2 × 35 oe]
			310% of the premium is £3875 The premium is $\frac{£3875 \times 100}{310}$	M1	3875 ÷ their 3.1 even if one term was missing from addition but must come from attempt at the **appropriate** addition [ie an error in adding to 310 or an omission of one term, aninclusion of say an extra 65, or an addition of 100, 70, 60,...etc with subsequent subtraction from 700]
			=£1250	A1	£1250 cao **Examiner's Comments** This was a good discriminator on Paper B. There were many good solutions scoring full marks. Others often included the wrong number of 0.65s in their total or sometimes incorrectly used 0.7+0.6+0.5+0.4.....and failed to subtract. Some others had no idea and tried to find say 65% of £3875.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
4			Basic premium = 35% of driver's premium	M1	use of 35% or 0.35 oe [not 65, 0.65 unless 1-0.65]
			Drivers premium = Basic premium ÷ 0.35 = 2.86 Basic premium $\Rightarrow k = 2.86$	A1	accept 2.86 or better (k = 2.85714… or $2\frac{6}{7}$ oe) [k = 2.9, k = 1/0.35 scores M1A0] **Examiner's Comments** This was not particularly well answered. Candidates seemed almost equally to find correctly that k=1/0.35= 2.86 or to use 1/0.65. A surprising number used 1-0.65 as 0.45.
			Total	2	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>5</td><td></td><td>i</td><td><table><tr><td>Group</td><td>P</td><td>Q</td><td>R</td><td>S</td></tr><tr><td>Number of people</td><td>15</td><td>30</td><td>60</td><td>45</td></tr><tr><td>Average number of accidents in a year</td><td>1.5</td><td>4.5</td><td>3</td><td>9</td></tr><tr><td>Average cost of accidents per year</td><td>£7500</td><td>£9000</td><td>£3000</td><td>£4500</td></tr></table></td><td>B1</td><td>for all three entries 30, 60, 45 correct</td></tr>
<tr><td></td><td></td><td>i</td><td></td><td>B1</td><td>for all three entries 1.5, 3, 9 correct</td></tr>
<tr><td></td><td></td><td>i</td><td></td><td>B1</td><td>for 3000, 4500 both correct</td></tr>
<tr><td></td><td></td><td>i</td><td></td><td></td><td>SC B2 for all entries in any three columns correct

Examiner's Comments

Most candidates correctly completed the table.</td></tr>
<tr><td></td><td></td><td>ii</td><td>£24000 ÷ 150</td><td>M1</td><td>Adding their bottom row (7500 + 9000 + ‘3000’ + ‘4500’ = ‘24000’) and dividing by 150 soi (and not divided or multiplied by any additional values)</td></tr>
<tr><td></td><td></td><td>ii</td><td>=£160</td><td>A1</td><td>ft their 24000÷150 (corr to 2dp if inexact)

Examiner's Comments

£24000/150=£160 was usually correct. Some candidates multiplied or divided by additional values, usually 18 or 9.</td></tr>
<tr><td></td><td></td><td>iii</td><td>The 45 members of Group S pay 1.5 × £4500 = £6750.
So each pays £6750÷45</td><td>M1</td><td>their 4500 × 1.5 oe soi as part of solution</td></tr>
<tr><td></td><td></td><td>iii</td><td>=£150.</td><td>A1</td><td>cao www (must be from correct final column)

Examiner's Comments

This was often correct but often long methods were seen including £24000x1.5=£36000, then 18.75% of £36000=£6750 instead of the more direct approach of £4500 × 1.5=£6750. Some candidates stopped at £6750.</td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
			Total	7	

Question			Answer/Indicative content	Marks	Guidance
6			n = 15, f = 6 gives 5.6	B1	5.6
			n = 15, f = 4 gives 3.9	B1	3.9
			so 'usually' 6 and 4 stops respectively	E1	Dependent on B1B1 – additional values of f considered E0 **Examiner's Comments** Part (i) was answered extremely well with nearly all candidates correctly expressing $\frac{1}{(1+2x)(1-x)}$ in partial fractions. In part (ii) the majority of candidates were able to separate the variables and substitute their partial fractions correctly. There were, however, frequent errors in the integration usually when candidates forgot to divide by 2 when integrating $\frac{1}{1+2x}$ and/or when they forgot to do the same process with the –l when integrating $\frac{1}{1-x}$. Many candidates did not include a constant of integration or, if included, it was either subsequently ignored or set to zero without any mathematical justification. Most candidates were able to combine their logarithmic terms correctly, though examiners noted the high volume of cases in which the 'correct' printed answer was seen following earlier incorrect working. Most candidates achieved the first two marks in part (iii) for finding the value of k , but many made errors in handling the logarithms to find the time taken for the drug concentration to reach 90% of its maximum value. Examiners noted the

Question			Answer/Indicative content	Marks	Guidance
					large variation in the accuracy of candidates' final answers in this part. In part (iv) most candidates multiplied up by l $-x$, collected and factorised the x terms correctly. The main problem seemed to be how to get the negative exponent. These often appeared when candidates divided e ^{3kt} –l by 2+e^{kt}, losing the final accuracy mark in the process. It was also common for candidates to simply not show how $\frac{e^{3kt}-1}{2+e^{3kt}}$ was equal to the given answer. Those candidates who started by taking the reciprocal of the answer given in part (ii) were usually far more successful in deriving the required result in this part. The majority of candidates who attempted to verify that the drug concentration approached its maximum value in the long term recognised that as $t \to \infty$, $e^{-kt} \to 0$, although some candidates simply substituted a large value of t to show that x was close to 1, this approach was not sufficient to earn the final mark in this part.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
7			$1-\left(\frac{9}{10}\right)^8 = 0.5695...$	M1	Substitution of the correct values into the correct formula
				A1	Accept 2 to 4 dp **Examiner's Comments** There seemed to be a misconception amongst a number of candidates about the definition of 'show' and 'verify' in this first part. The majority of candidates correctly substituted the coordinates of A, B and E into the equation of the plane, to achieve the first two marks. Those candidates who tried to derive the equation of the plane had methods that were generally incomplete and despite a lot of work, did not gain much credit. In part (ii)(A) a number of candidates seemed to think that just showing one direction vector was normal to the plane was sufficient and some candidates showed all three. It was unfortunate that so many lost marks by not showing the evaluation of the scalar product(s) even though this was another 'show that' question and so examiners had to be convinced that the candidates were indeed showing the required results and not simply stating them. Part (ii)(B) was nearly always done correctly with only a small minority having the equation of the plane as x –6y–6=0rather than the correct x –6y+6=0. Part (ii)(C) occasionally contained simple arithmetical errors with the correct equation often leading to either b = $b=\frac{3}{4}$ or b = – $b=-\frac{4}{3}$. Some candidates stated the direction vector $\overrightarrow{FG}$ correctly and then either did not state the length of FG or resorted to using Pythagoras to calculate

Question			Answer/Indicative content	Marks	Guidance
					this length, not recognising that they could simply read off the required answer. To find the angle EFB in part (iii) most candidates attempted to find vectors (and hence use the scalar product) that would lead to the correct angle, though there were common arithmetical errors here. Some candidates used $\overrightarrow{EF} \bullet \overrightarrow{FB}$ and hence obtained an acute angle, often not recognising that they needed to subtract this from 180 as angle EFB was obtuse. Some candidates who used $\overrightarrow{FE}$ and $\overrightarrow{FB}$ 'lost' the minus sign in their calculation and ended up with an answer of 36.9°. A very small number of candidates calculated the lengths of the three sides of triangle EFB and used the cosine rule. Part (iv) required candidates to find the height of Q above the ground. Those candidates who thought the *y* – axis represented the vertical height clearly did not look at the diagram carefully enough. A surprising number of candidates could not calculate the *z* coordinate of P correctly. Some candidates seemed to think that in finding $\overrightarrow{PQ}$ they had arrived at the required answer whereas other candidates worked out the direct distance from the origin to point Q. Too many candidates worked out both the *x* and *y* coordinates of Q when only the *z* coordinate was required. Finally, a number of candidates found the correct position vector for Q but did not explicitly state the height of Q above the ground.
			Total	2	

<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>8</td><td></td><td>i</td><td>
<table>
<tr><td></td><td colspan="2">Car C</td></tr>
<tr><td></td><td>Arrival time (seconds)</td><td>Departure time (seconds)</td></tr>
<tr><td>Ground floor</td><td>0</td><td>20</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>Floor 7</td><td>41</td><td>56</td></tr>
<tr><td>Floor 8</td><td>59</td><td>74</td></tr>
<tr><td></td><td></td><td></td></tr>
<tr><td>Return to ground floor</td><td>98</td><td></td></tr>
</table>
</td><td>B1</td><td></td></tr>
<tr><td></td><td></td><td>i</td><td></td><td>B1</td><td>59, 74 or their 56 + 3, their 59 + 15</td></tr>
<tr><td></td><td></td><td>i</td><td></td><td>B1</td><td>98 or their 74 + 24</td></tr>
<tr><td></td><td></td><td>ii</td><td>4×98s = 392s (6 mins 32s)</td><td>B1</td><td>cao

Examiner's Comments

In part (i) the majority of candidates correctly wrote down the result that $\frac{dy}{dx}=\frac{2\sec^2\theta}{\sec\theta\tan\theta}$. It was clear though that a significant number of candidates did not know that $\frac{d}{d\theta}(\tan\theta)=\sec^2\theta$, and the majority of these applied the quotient rule to $\frac{\sin\theta}{\cos\theta}$ in an attempt to derive this result. A significant minority of candidates did not show adequate working to gain the final mark in part (i) with many going directly from $\frac{2\sec\theta}{\tan\theta}$ (or equivalent) to the given answer. Candidates are reminded that in questions in which the demand is to show a given result then sufficient working must be shown to satisfy the examiner that</td></tr>
</table>

Question			Answer/Indicative content	Marks	Guidance
					the answer has been correctly derived and not simply written down. It was also noted by examiners that a number of candidates multiplied their expressions for $\frac{dy}{d\theta}$ and $\frac{dx}{d\theta}$ rather than dividing. Although the techniques of verifying the Cartesian equation of the curve was relatively straightforward, answers in part (ii) were poorly presented, with many candidates unable to present a mathematical justification clearly and formally. Some candidates were insufficiently confident in the use of standard trigonometric identities and for some candidates it was as if they had never seen the relationship that $1+\tan^2\theta \equiv \sec^2\theta$. In part (iii) the vast majority of candidates considered both the correct integral for the volume of revolution and integrated $4x^2 - 4$ correctly. It was the mention of a rotation of 180 that seemed to concern many candidates and a considerable number divided the correct answer of $1+\tan^2\theta \equiv \sec^2\theta$ by 2.
			Total	4	

Question			Answer/Indicative content	Marks	Guidance
9				B1*	Correct calculation eg 164 + 24 or 230 – (30 + 12) (oe) but not 200 – 12
			Lift calls at floors 1 to 8 (only), leaving floor 8 at 164 seconds. Descent takes 24 seconds	E1dep*	Correct justification of the correct calculation (dependent on B1). If the departure time of 164 is used then their explanation must imply the use of the first 8 floors (not just '8 floors') **Examiner's Comments** This question differentiated well due to the coefficient of sin *x* taking the form of a positive constant rather than a number. Many candidates, however, were unfazed by this and worked out the correct values for *R* and α. Some candidates lost the first method mark by not including *R* in the expanded trigonometric statements Rcos α = 1,*R* sin α=λ. Writing α in terms of the more complex arcsin and arccos expressions was surprisingly common. It was a little worrying that a sizeable minority of candidates went from the correct $R=\sqrt{1+\lambda^2}$ to the incorrect *R* =1 + λ, thinking the squared terms and the square root cancelled each other out. In part (ii) those candidates that realised that *R*=2 usually went on to get the correct values for λ and α. However it was common for λ to be incorrect due to an incorrect expression for *R* from part (i). A fair proportion of candidates gave α in degrees and those who gave α as either arccos$\left(\frac{1}{\sqrt{1+\lambda^2}}\right)$ or arcsin$\left(\frac{\lambda}{\sqrt{1+\lambda^2}}\right)$ were generally less successful in this part than those who gave α as arctan λ.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
10			$10\left(1-\left(\frac{9}{10}\right)^7\right)=5.22$	M1	Substitution of the correct values into the correct formula
				A1	5.22 (must be 2 dp), B2 5.22 www (isw after correct answer seen) **Examiner's Comments** The most common mistake in part (i) was to use a value of 2 rather than –2 as the coefficient of *x* in each term of the expansion. The binomial coefficients were nearly always correct though a small number missed the 2! from the denominator of the x^2 term. While the majority of candidates used the correct value of n a small minority incorrectly used $n = n = \frac{1}{3}$ or $-\frac{2}{3}$. The range of validity of the expansion was done much better than in previous years although the most common mistake was to give non-strict inequalities. Other mistakes included: $-\frac{1}{2} < -x < \frac{1}{2}$ $\lvert x \rvert > \frac{1}{2}$ $\frac{1}{2} < x < -\frac{1}{2}$ In part (ii) the majority of candidates correctly multiplied their answer from part (i) with (1–3*x*) and simplified this expression correctly to obtain the correct values of a and b . It was concerning, however, that a number of candidates wrote $(1-3x)(1+\frac{2}{3}x+\frac{8}{9}x^2+\ldots)=1+\frac{2}{3}x+\frac{8}{9}x^2-3x-2x-\frac{8}{3}x^2+\ldots$ or even more worryingly expanded $(1-3x)^1$ as 1 – 3*x* + higher order terms in *x*.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
11			$\frac{110}{146} \times 100\% = 75.3\%$	M1	M1 allow (90 or 95 or 125) / 146 × 100
				A1	75.3 or better (eg 75.34 etc), 75 is A0 B2 75.3 with no working, B1 0.753 (oe) **Examiner's Comments** The majority of candidates correctly replaced cos 2θ with 1 – 2 $\sin^2θ$ although a minority of candidates made the costly mistake of replacing cos2θ with 1 –sin $^2θ$. While some candidates struggled to factorise 12sin $^2θ$ – sinθ –6=0 many used the quadratic formula to solve this equation, and as with question 1, there were some candidates who did not state or apply the quadratic formula correctly. While the majority of candidates found the correct values for sinθ some incorrectly obtained – $-\frac{3}{4}$ and/or $\frac{2}{3}$. Of those candidates that obtained the correct values for sinθ the majority went on to score full marks. However, it was fairly common to see 'arcsin $\left(-\frac{2}{3}\right)$ = –41.81… *therefore no solutions in the range*' with no appreciation that solutions in the correct range could be found from this value. Having found the principal values it was common for candidates to get the other solutions in the range, often sketching the sine curve to help them, though most did this correctly without demonstrating any method.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
12		i	E2. Pick up Q	B1	
		ii	E5. Drop off Q	B1	
		iii	E4. Drop off P and R	B1	**Examiner's Comments** Common errors included: • $5(x+1)–3(2x+1)$ (and not the correct $5x(x+1)–\ldots$) • Expanding $–3(2x+1)$ as either $–6x+3$ or $–6x+1$ • There were some candidates who did not multiply up on the right-hand side and so obtained $5x(x+1) –3(2x+1)=1$ • Some lost the final two marks for not applying the quadratic formula correctly However, this question was generally done well with most candidates scoring full marks and demonstrating sound basic algebraic manipulation skills. It was common to see the use of the quadratic formula as much as factorising to solve the final quadratic equation. Very few completed the square but those that did were mainly successful.
			Total	3	